Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/02/2004
PI Name:	Sayler, Gary Ph.D.		
Project Title:	Bioluminescent Monitoring of Opportunistic Pathogens in the Spacecraft Environment		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	37996	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-01
Start Date:	12/31/2002	**End Date:**	01/01/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ripp, Steven (The University of Tennessee) Simpson, Michael L (Oak Ridge National Laboratory)		
Grant/Contract No.:	NAG9-1424		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The specific research objective of this proposal is to insert a quorum sensing gene (luxI) into bacteriophage to intrinsically generate autoinducer molecules upon target cell infection and subsequent amplification of lux-based bioluminescent signaling mechanisms. Using quorum sensing molecules as a means of detection allows for sensing of biological agents at very low level concentrations. As opposed to chemical detection systems, where typical toxic thresholds of part-per-billion to part-per-million still equate to >1014 molecules available for detection, a quorum sensing biological detection system may be capable of sensing down to one individual biological agent. This technology provides a unique mechanism to provide a measurable amount of bioluminescence from these low event numbers by using the production of autoinducer molecules to produce amplification of individual targeted biological agents, thus avoiding dependence upon target cell growth. To develop microelectronic-based bacteriophage-amplified bioluminescent biosensors for the specific detection and identification of biological contaminants in the spacecraft		

	environment. The sensors are capable of functioning as a network of autonomous detectors for continuous on-line, near real-time monitoring in a low-power, low-mass, low-maintenance, redundant framework.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ability to rapidly detect target pathogens has earth benefits in medical diagnostics, food safety, environmental monitoring, and homeland defense.
Task Progress:	The premise of this research plan was to genetically engineer bacteriophage such that they would generate a chemical signature upon infection of a host specific pathogen. Thus, rather than attempting the difficult task of detecting pathogens at low numbers, we can use phage infection to convert each pathogen into an amplified chemical signature, and then use bioreporters sensitive to the chemical signature to report its presence through bioluminescent signaling. We have initially proven this technology using Escherichia coli as a model host organism. Bacteriophage lambda was engineered to contain the luxI gene responsible for synthesis of the autoinducer molecule N-3-(oxohexanoyl)-L-homoserine lactone (OHHL). In addition, a bioreporter was constructed to respond to OHHL molecules through bioluminescent signaling (using the luxCDABE cassette of Vibrio fischeri). The assay requires only the combination of bacteriophage, bioreporter, and E. coli cells, which was performed in 96-well microtiter plates with bioluminescent responses detected automatically in a Perkin-Elmer Victor2 Multilabel reader. Significant self-generated bioluminescent responses could be detected from 1 to 1 × 104 E. coli cells within 8 – 11 hours post-inoculation. At cell densities >104, detection could be achieved within 2 – 7 hours. We are currently trying to increase the sensitivity and rapidity of the assay by incorporating multiple copies of luxI into the phage, rather than the single copy now present. This should result in elevated synthesis of OHHL, and, consequently, detection of fewer E. coli in less time. Eventually, we also aim to replace the bulky multilabel reader with an integrated circuit luminometer, thereby allowing on-chip sensing of the bioluminescent signal in a standalone, miniaturized format. The luminometer chip is currently being upgraded to include wireless transmission and is undergoing some redesign to ‘ruggedize’ it for eventual space application. Other bacteriophage assays are also being developed for Pseudomonas aeruginosa, Salmonella, and Staphylococcus aureus.
Bibliography Type:	Description: (Last Updated: 03/16/2017)
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Patents	6,544,729. Complete, April 2003. Apr-2003 Sayler, G.S.; Ripp, S., Applegate, B. "Bioluminescent biosensor device."